Warning and Detection of Forest Fire

A Project Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

Information Technology

by

Vismaya K M (IDK20IT047)

Shegin Ismail A P (IDK20IT038)

Sreya Subramanian (IDK20IT040)

Vishnu C N (IDK20IT046)



DEPARTMENT OF INFORMATION TECHNOLOGY

GOVERNMENT ENGINEERING COLLEGE IDUKKI, PAINAVU - 685603

KERALA

June 2023

DEPARTMENT OF INFORMATION TECHNOLOGY

GOVERNMENT ENGINEERING COLLEGE IDUKKI, PAINAVU-685603

2022 - 23

CERTIFICATE

This is to certify that the report entitled **Warning and Detection of Forest Fire** submitted by **Vismaya K M** (IDK20IT047), **Shegin Ismail A P** (IDK20IT038), **Sreya Subramanian** (IDK20IT040) & **Vishnu C N** (IDK20IT046) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Information Technology is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Dr. Deepthi Sasidharan
(Project Guide)
Associate Professor & Head
Dept. of IT, GECI

Prof. Sanu C
(Project Coordinator)
Assistant Professor
Dept. of IT, GECI

Dr. Visakh R
(Project Coordinator)
Assistant Professor
Dept. of IT, GECI

Prof. Sonia Joseph
(Project Coordinator)
Assistant Professor
Dept. of IT, GECI

Prof. Keerthimol P P
(Project Coordinator)
Assistant Professor
Dept. of IT, GECI

DECLARATION

We hereby declare that the project report **Warning and Detection of Forest Fire**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under the supervision of Dr. Deepthi Sasidharan.

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Vismaya K M

Shegin Ismail A P

Sreya Subramanian

Vishnu C N

Painavu

18-06-2023

Abstract

Forest fires pose a significant threat to ecosystems, wildlife, and human lives. In this project we aims to develop a forest fire warning and detection system using image processing techniques, OpenCV, and IoT sensors with temperature and smoke detection capabilities. The system analyzes real-time images captured by surveillance cameras, enabling visual patterns, color detection, and smoke detection algorithms to identify potential fire occurrences. IoT devices with smoke and temperature sensors are strategically placed throughout the forest to continuously monitor environmental conditions. The system uses advanced computer vision techniques, websockets, and Flutter to facilitate real-time monitoring and interface.

The web app displays sensor data, allowing users to remotely monitor temperature levels and smoke presence. In case of abnormal readings, the app triggers alerts and notifications, enabling swift intervention by authorities. This proactive approach to forest fire detection minimizes damage, safeguards wildlife, and protects human lives.

Acknowledgement

It is our privilege to express our sincere and deep gratitude and indebtedness to the people who helped us to complete the project successfully. All glory and honors to Almighty God, who showered His grace upon us to make this endeavor a success. We thank Dr. Suresh K., Principal, Government Engineering College Idukki, for providing new dimensions to this engineering course. We express our gratitude to Dr. Deepthi Sasidharan, Head and project guide, Department of Information Technology, Government Engineering College Idukki, for her valuable suggestions, encouragement, and advice for the fruitful completion of this project.

We express our deep sense of gratitude to the project coordinators Prof. Sanu C, Dr. Visakh R, Prof. Sonia Joseph, and Prof. Keerthimol P P, Assistant Professors in the Department of Information Technology, Government Engineering College Idukki, for their valuable suggestions and review comments throughout the project work. Thanks are also due to all the teaching and non-teaching staff of the Department of Information Technology, Government Engineering College Idukki, for their valuable support and help. We also thank our beloved friends for their timely help in peer reviewing our reports and valuable suggestions to improve our work to perfection.

Vismaya K M

Shegin Ismail A P

Sreya Subramanian

Vishnu C N

Contents

Abstract i

Acknowledgement ii

List of Figures v

List of Tables vi

1 Introduction 1

2 Literature Review 2

2.1 A Survey on Existing Fire Detection Techniques 2

2.1.1 Fire Detection Technique Using Video Capture and Processing 2

2.1.2 Fire Detection Using Sensors 5

2.2 Outcome of the Survey 7

3 Problem Definition 9

3.1 Problem Statement 9

3.2 Problem Description 9

3.3 Scope 9

3.4 Objectives 10

4 System Development 12

4.1 Software Requirements 12

4.1.1 Requirement Specification 12

4.1.2 Functional Requirements 12

4.1.3 Non-Functional Requirements 14

4.2 Hardware Requirements 14
4.2.1 Introduction 14
4.2.2 Hardware Components 15

5 System Design **16**
5.1 System Architecture 16
5.2 Usecase Diagram 18
5.3 Class Diagram 19

6 Implementation **20**
6.1 Tools Used for Designing Web Application Interface 20
6.1.1 Flutter 20
6.1.2 Android Studio 21
6.1.3 Visual Studio Code 21
6.2 Tools Used for Designing Hardware Interface 21
6.2.1 Android IDE: 21
6.2.2 ESP32 Board: 22
6.2.3 ThingSpeak: 22
6.3 Tools Used for Image Processsing 22
6.3.1 Python: 22

7 Results and Discussion **23**
7.1 Results and Discussion 23
7.2 Screenshots 24

8 Testing **31**
8.1 Functionality Testing 31
8.2 UI Testing 31
8.3 Integration Testing 32
8.4 Network Testing 32

9 Conclusion **34**

References **35**

List of Figures

2.1 Flowchart for Fire Detection Using GMM Techniques 5
2.2 Flowchart for Fire Detection Using Temperature and Gas Sensor . . . 6

5.1 Flow Chart 16
5.2 System Architecture 17
5.3 Usecase Diagram 18
5.4 Class Diagram 19

7.1 login page 25
7.2 Home Page 25
7.3 Location Page 26
7.4 Statistics Page 26
7.5 Livefeed from IP Camera 27
7.6 Mobile IP Camera 27
7.7 Fire Detected From IP Stream 28
7.8 Hardware Components 28
7.9 Smoke initiated 29
7.10 Fire Detected From IP Stream and Smoke Sensor 29
7.11 Log Page 30

List of Tables

2.1 Performance Comparison 4
2.2 Summary of Literature Review 8

Chapter 1

Introduction

The project combines image processing with IoT technologies to develop an efficient forest fire detection and warning system. By integrating OpenCV and Python's Haar Cascade Classifier, our system can accurately identify fires in real-time video frames through visual pattern analysis, color detection, and smoke detection algorithms. To enhance detection capabilities, IoT devices equipped with smoke and temperature sensors are strategically placed in forested areas. These sensors continuously monitor environmental conditions and transmit real-time data to our system when thresholds are exceeded. A dedicated web application integrates livestream frame data and IoT sensor feedback, generating alarms and notifications for prompt intervention.

The comprehensive solution addresses the urgent need for proactive forest fire detection and response. By leveraging image processing and IoT technologies, our system minimizes the impact of wildfires on ecosystems, wildlife, and human lives. The technical foundation of the project relies on OpenCV and Python for fire detection, while IoT devices enable real-time data transmission. The web application offers an interface for monitoring and generating timely alerts. This project report provides an overview of our solution, emphasizing its technical aspects and potential benefits in the field of fire detection and response.

Chapter 2

Literature Review

Fire detection systems are essential for ensuring the safety of lives and properties by providing early warnings and facilitating rapid responses to fire incidents. As fire hazards continue to pose significant threats, advancements in fire detection technologies have garnered increased attention. This literature review aims to analyze scholarly articles, research papers, and industry reports to identify trends, advancements, challenges, and gaps in the field of fire detection systems. By synthesizing existing literature, this review aims to inform future developments and enhance fire safety practices.

2.1 A Survey on Existing Fire Detection Techniques

The Survey regarding the Existing Fire Detection Technique was conducted based on the Primary information provided by J. Marin K *et al.*[1]. Some of the significant findings related to the Detection Technique are stated below

2.1.1 Fire Detection Technique Using Video Capture and Processing

Fire Detection Technique using Robust-Adaboost

The paper by X. Wu and H. Leung *et al.*[2], proposes a novel method for developing a fire smoke detection system based on video analysis using a robust version of the

AdaBoost algorithm. The goal of their approach is to enhance the accuracy and robustness of existing fire detection methods.

The robust AdaBoost algorithm is employed to handle complex and changing visual conditions by selecting and combining multiple weak classifiers to form a strong classifier. This approach allows the system to adapt to various smoke and fire patterns, thereby improving the overall detection performance.

The general steps of the robust AdaBoost algorithm, as mentioned in the paper, can be summarized as follow Initialization: Initialize the weights of the training samples.

For each iteration:

- Train a weak classifier: Select a weak classifier that performs better than random guessing on the training samples.
- Compute the error rate: Evaluate the performance of the weak classifier by calculating the weighted error rate.
- Update sample weights: Adjust the weights of the training samples based on their classification errors.
- Normalize sample weights: Normalize the updated weights so that they sum up to 1.
- Adjust weak classifier weight: Assign a weight to the weak classifier based on its error rate.

  Combine weak classifiers: Assign weights to each weak classifier and combine them to form a strong classifier.

  Classification: Utilize the final strong classifier to classify new, unseen video sequences based on their weighted votes.

Accuracy of the proposed method with different feature combinations, [29,34] are given in the Table 2.1

Table 2.1: Performance Comparison

Method	Accuracy (%)	Detection Rate (%)	False Alarm Rate (%)
Cai et al. [29]	74.10	92.70	7.30
Wu et al. [34]	73.41	94.48	5.52
AB-RBFSVM+ (with CES+MD+MS)	88.28	99.52	0.48
AB-RBFSVM+ (with TCW+MDC+MS)	91.25	99.69	0.31

Video Fire Detection Based on Gaussian Mixture Model and Multi Color Features

The paper by X. Han *et al.*[3] presents a video fire detection method using the Gaussian Mixture Model (GMM) and multi-color features. Their approach involves the following steps:

- Background Modeling: GMM is used to model the background by capturing statistical properties from non-fire frames.
- Fire Pixel Detection: Color features of each pixel are compared with the background model to detect fire pixels.
- Spatial and Temporal Filtering: Techniques are applied to reduce false alarms by considering neighboring pixels and analyzing consecutive frames.
- Fire Region Segmentation: Fire pixels are grouped together to form coherent fire regions using connected component analysis and region merging.
- Fire Verification: Shape, texture, and motion characteristics of the fire regions are analyzed to validate and minimize false positives.

By combining GMM, multi-color features, spatial and temporal filtering, and fire region segmentation, their method achieves accurate fire detection in video sequences while reducing false alarms. The detailed flowchart of how GMM works is illustrated in the Figure 2.1.

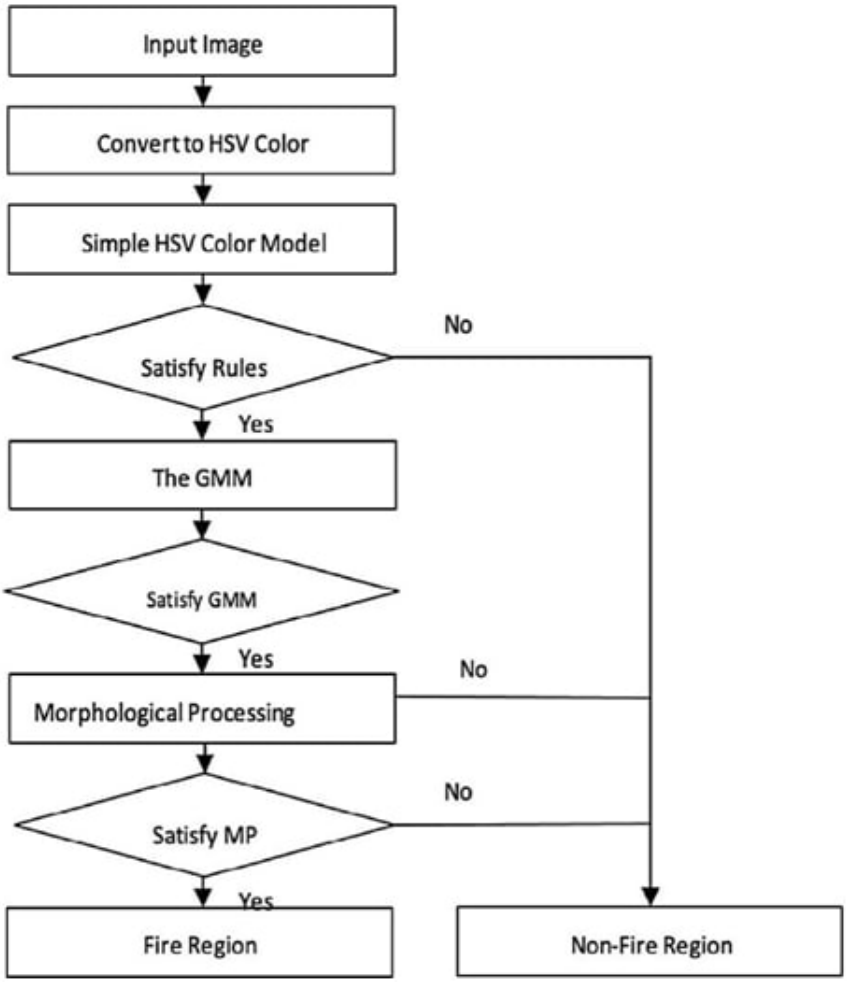


Figure 2.1: Flowchart for Fire Detection Using GMM Techniques

2.1.2 Fire Detection Using Sensors

The paper by F.Khan *et al.*[4] Provides an introduction to the importance of fire detection and the role of sensors in fire safety systems and highlights the need for advanced sensors to improve fire detection accuracy and response time. authors discuss traditional fire detection sensors, including heat sensors and smoke sensors. Here's a brief overview of these sensors Heat and smoke sensors enhance the efficiency of fire detection systems through their unique capabilities:

- Heat sensors provide early fire detection by sensing temperature changes, enabling prompt response measures. They are specific to fire-related temperature increases, minimizing false alarms. Compatible with various fire detection systems, they contribute to seamless integration and redundancy with other

sensors.

- Smoke sensors offer early fire detection by sensing smoke particles, even before the presence of flames. They are highly sensitive and can detect minute amounts of smoke, ensuring early warning and intervention. Rapid alarm activation facilitates swift response, reducing the fire spread and damage. Strategically placed sensors provide comprehensive coverage, even in concealed spaces. Advanced smoke sensors employ algorithms to reduce false alarms caused by non-fire sources, maintaining system reliability.

  The combination of heat and smoke sensors enhances accuracy, early warning, and reliability in fire detection systems. This integration enables timely response, minimizes damage, and increases overall safety.The working of fire detection using sensors is illustrated as flowchart in the Figure 2.2 below.

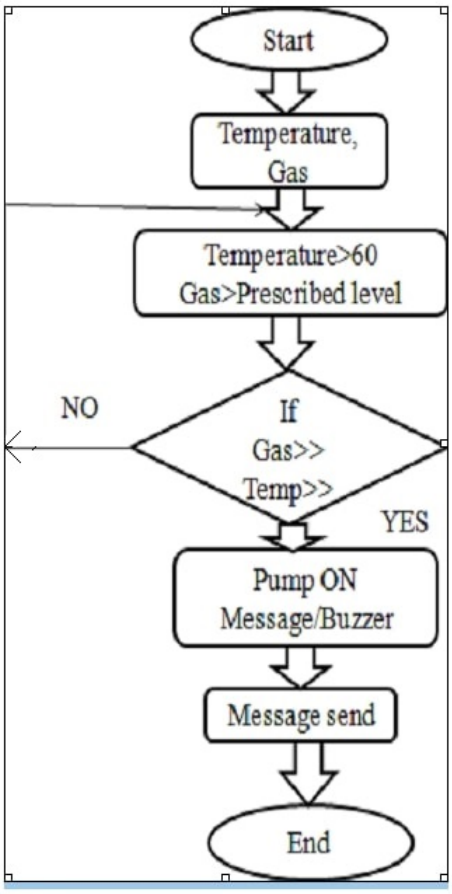


Figure 2.2: Flowchart for Fire Detection Using Temperature and Gas Sensor

2.2 Outcome of the Survey

The conclusion drawn from the survey conducted on various fire detection techniques was that each method had its own pros and cons. To address this, a decision was made to integrate multiple systems, including detection using image processing and IoT implementation of sensors.

In the image processing sector, the Robust AdaBoost algorithm was chosen as the method to train the model. This algorithm combines several weak classifiers to develop a strong algorithm with reduced false detections. Instead of manually training the classes with images, an alternative approach was adopted based on the Gaussian Mixture Model and Multi-Color Features. This technique extracts the HSV color model from the pixels in the pictures, allowing for the identification and characterization of fire in the images.

To implement the image processing and IoT integration, Python's computer vision technology using OpenCV and Flask was utilized based on the findings from S. Mehtab and J. Sen[5]. OpenCV provides effective tools for image analysis and processing, while Flask facilitates the development of a web application. Real-time data from temperature and smoke sensors is collected to obtain accurate temperature and smoke values.

The combination of image processing and IoT implementation forms a web application that alerts the Forest Authority in real-time about the presence of fire. This integration ensures timely and efficient fire detection, enabling prompt actions to be taken to mitigate the fire risk. The web application serves as a valuable tool for the Forest Authority to monitor and respond to fire incidents effectively. The summary of the Literature Review is stated in the Table 2.2 as follows.

NB: the first reference paper by J. Marin K et al.[1] was utilized to identify various means of fire detection strategies, it does not provide exhaustive details on the methods employed. Therefore, it is not included in the Table 2.2 .

Table 2.2: Summary of Literature Review

Authors	**Technique Used**	**Advantages**	**Disadvantages**
Wu, X. et al.[2]	Video-based fire and smoke detection using Robust AdaBoost.	Robust detection algorithm, high accuracy, Real-time Detection.	Dependency on video input quality, Limited to Video Input i.e., not applicable to static images or sensor readings.
X. Han et al.[3]	Video fire detection based on Gaussian Mixture Model and multi-color features.	Effective fire detection using color analysis, Minimal False Alarms.	Sensitive to lighting conditions, Computational Complexity.
F. Khan et al.[4]	Recent advances in sensors for fire detection and Overview of advancements in sensor technology.	Remote and Continous monitoring, Early detection.	Maintenance and calibration, Limited detection range.
S. Mehtab and J. Sen[5]	Object detection and tracking using OpenCV in Python.	Cost-effective, Flexibility and adaptability, Real-time monitoring.	Reliance on accurate object detection.

Chapter 3

Problem Definition

The chapter defines the problem statement, problem description, scope, and objectives of our system.

3.1 Problem Statement

To develop an warning system that detects forest fire using Image Processing and IOT

3.2 Problem Description

Forest fires pose a significant threat to natural ecosystems, wildlife, and human lives. Detecting forest fires in a timely manner is crucial for preventing extensive damage. However, existing fire detection methods may have limitations in terms of coverage, accuracy, and real-time monitoring. Therefore, the project aims to develop a warning system that utilizes image processing and IoT technology to detect forest fires and provide timely alerts to the authorities.

3.3 Scope

In the present era,the integration of image processing and IoT technologies has revolutionized forest fire detection. Our system utilizes advanced tools like OpenCV and Python's Haar Cascade Classifier for exceptional accuracy in identifying fires in

real-time video frames. It combines visual pattern analysis, color detection, and smoke detection algorithms.

Strategically positioned IoT devices with smoke and temperature sensors monitor environmental conditions. They transmit real-time data to our system when thresholds are exceeded. The web application integrates livestream frame data and IoT sensor feedback, generating alarms for timely intervention.

Our comprehensive solution proactively detects and mitigates the impact of forest fires. By leveraging image processing and IoT technologies, it minimizes damage to ecosystems, wildlife, and human lives. OpenCV and Python provide the technical foundation for fire detection, while IoT devices enable real-time data transmission.

The web application offers an intuitive interface for monitoring and generating timely alerts. This project report highlights the technical aspects and potential benefits of our solution in the field of fire detection and response.

3.4 Objectives

The project has the following objectives:

1. Forest Fire Detection from Livestream: Develop algorithms and techniques to accurately analyze the livestream video from the IP camera. This involves utilizing image processing and computer vision methods to identify fire-related patterns and anomalies, enabling reliable detection of forest fires.

2. Forest Fire Detection using IoT Sensors: Implement an IoT network with strategically placed sensors to collect environmental data. Develop algorithms to analyze the collected data and identify abnormal patterns that indicate the presence of a potential forest fire. Statistical analysis and anomaly detection techniques will be employed to improve the accuracy of fire detection.

3. Data Analysis App Development: Create a web application that can receive and analyze data from the livestream and IoT sensors. Implement machine learning algorithms to process the data in real-time and provide immediate fire detection. The application will generate alerts and enabling authorities to take prompt action.

4. Authorities' Warning System: Develop a robust communication system between the web application and the authorities responsible for monitoring forest fires.

Implement mechanisms to ensure the timely delivery of alerts. Design and integrate an alarm buzzer that can be triggered by the application to prompt the authorities at the right time, facilitating immediate response to potential forest fires.

By achieving these objectives, the project aims to provide an efficient forest fire warning system that combines image processing, IoT technology, and web application analysis. The system will enhance the authorities' capabilities to detect and respond to forest fires promptly, thereby minimizing potential damage and protecting the forest ecosystem.

Chapter 4

System Development

The chapter is to specify the software and hardware requirements of the system.It serves as the foundation for the development of the software providing a detailed description of what the software should do and how it should behave.

4.1 Software Requirements

4.1.1 Requirement Specification

Introduction

The purpose of this software requirement specification is to outline the functional and non-functional requirements for the development of a project report generation software based on the provided code. The software aims to generate a comprehensive report based on the forest fire detection system's data and analysis. It will gather information from various sources, including the Thingspeak platform and a local server, to create an informative report.

4.1.2 Functional Requirements

Data Retrieval

- The software shall fetch real-time smoke and temperature data from a specific channel on the Thingspeak platform.

- The software shall establish an HTTP connection to retrieve the latest data at regular intervals.
- The software shall handle successful and failed data retrieval responses appropriately.

Alarm and Alert System

- The software shall trigger an alarm sound and display an alert message if the smoke value exceeds a predefined threshold.
- The software shall provide options to customize the alarm sound and threshold values.

Livestream Image Processing

- The software shall establish a WebSocket connection with a local server to receive detection responses.
- The software shall parse the received JSON-encoded detection information, including the detection status, description, and timestamp.
- The software shall display the detection information on the user interface in real-time.
- The software shall decode and display the base64-encoded frame as an image representing the detected area.

User Interface

- The software shall provide a user interface with a dedicated screen for the fire detection system.
- The user interface shall display real-time temperature and smoke readings.
- The user interface shall include buttons to start and stop the fire detection process.
- The user interface shall show an alert message when smoke exceeds the threshold.

- The user interface shall display the livestream detection information, including the detection status, description, timestamp, and visual representation of the detected area.

4.1.3 Non-Functional Requirements

Performance

- The software shall be responsive and provide real-time updates of temperature, smoke, and detection information.
- The software shall handle multiple concurrent user requests without significant performance degradation.

Reliability

- The software shall handle network disruptions and recover gracefully upon reconnection.
- The software shall be robust and able to handle unexpected input or invalid data.

Security The software shall ensure secure communication with external platforms and servers, including encryption of sensitive data. Usability

- The user interface shall be intuitive, visually appealing, and easy to navigate.
- The software shall provide appropriate error messages and notifications to guide users in case of any issues.

4.2 Hardware Requirements

4.2.1 Introduction

The purpose of this hardware requirement specification is to outline the necessary hardware components and their specifications for the implementation of the forest fire detection system based on the provided Arduino code. The system utilizes various sensors, microcontrollers, and other peripherals to monitor temperature, smoke levels, and transmit data to external platforms.

4.2.2 Hardware Components

ESP32 Development Board

- Compatible with the Arduino IDE and programming language.
- Supports the necessary digital and analog input/output pins.
- Provides built-in Wi-Fi capabilities.

MAX6675 Thermocouple Sensor Module

- Interfaced with the ESP8266 board using SPI communication.
- Measures temperature using a thermocouple probe.
- Provides accurate temperature readings.

MQ2 Smoke Sensor

- Interfaced with the ESP8266 board using analog input.
- Detects smoke or particles in the air.
- Provides an analog output proportional to smoke concentration.

ESP8266 Development Board

- Compatible with the Arduino IDE and programming language.
- Supports the necessary digital and analog input/output pins.
- Provides built-in Wi-Fi capabilities.

Buzzer

- Connected to appropriate digital output pins on the ESP32 board.
- Used to indicate alarm or alert conditions.

Chapter 5

System Design

This chapter defines the database design, system architecture, usecase diagram and class diagram of our system.

5.1 System Architecture

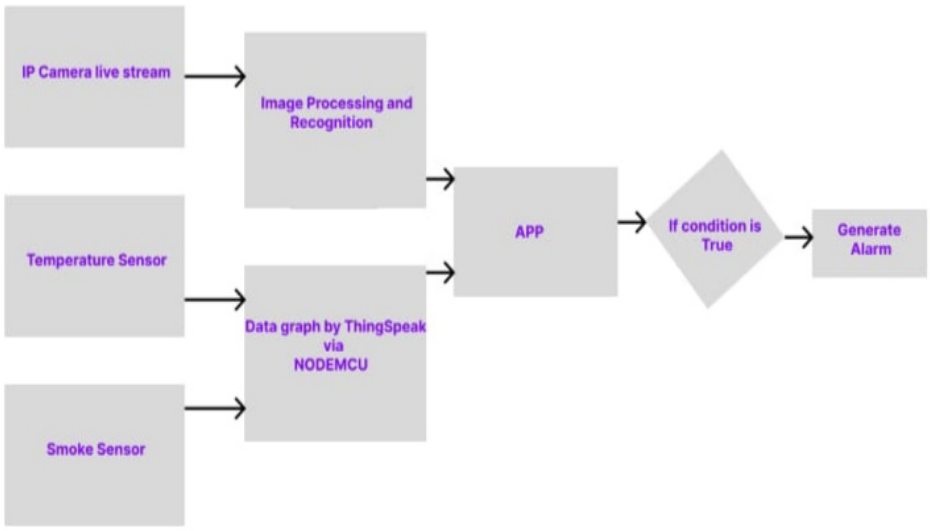


Figure 5.1: Flow Chart

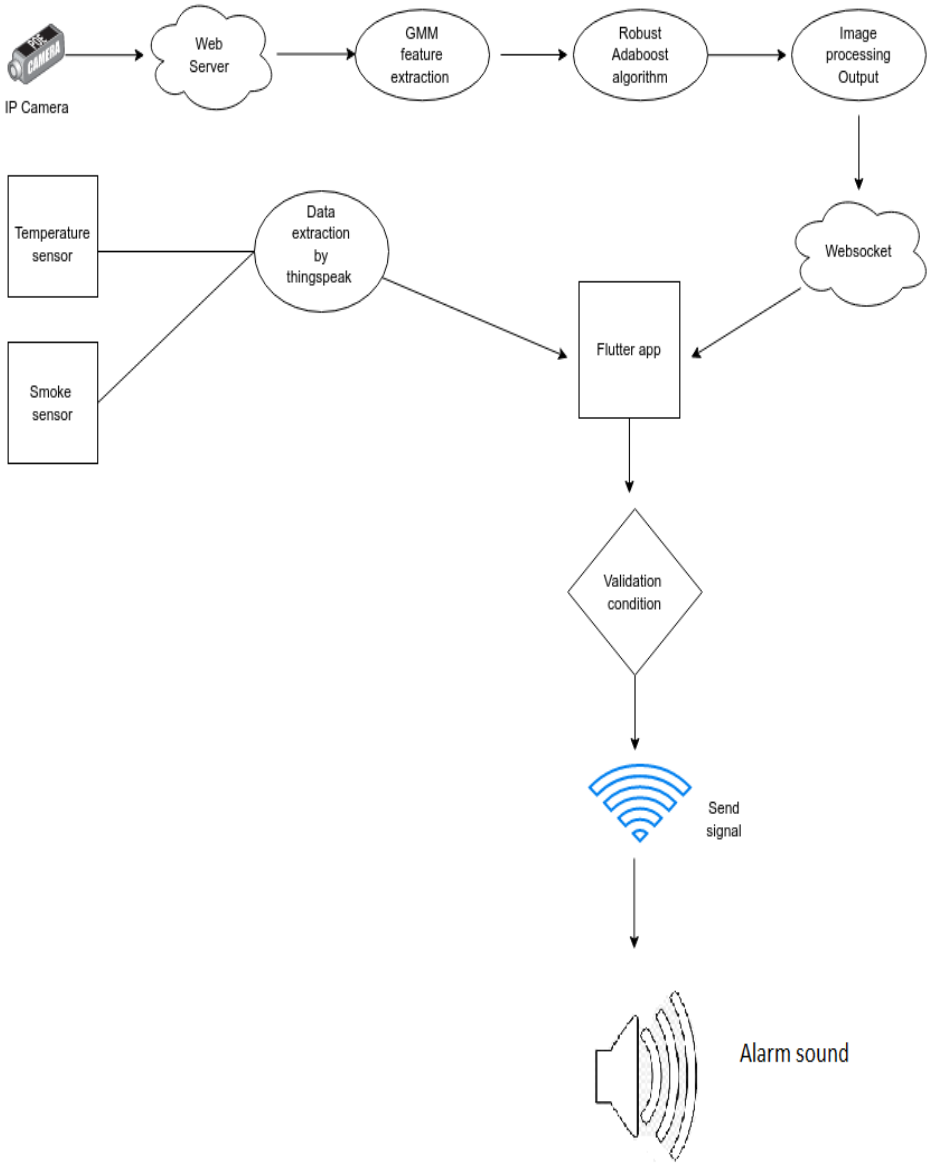


Figure 5.2: System Architecture

The Figure 5.2 shows the description of the system architecture.It is the high-level design of the system which outlines the overall structure and components of a system. It provides an abstract view of how different modules or components interact with each other to achieve the system's functionality. The Figure 5.1 depicts the block diagram of how the system actually gets the input and output feed and how they are composed to give the final output.

5.2 Usecase Diagram

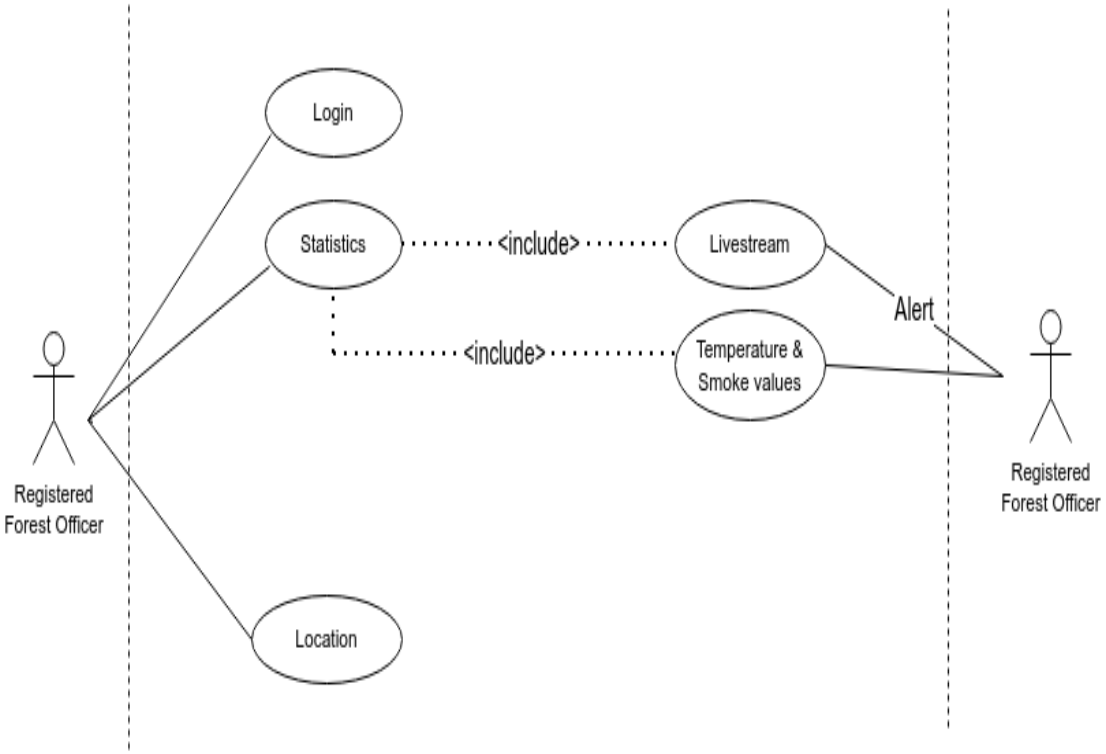
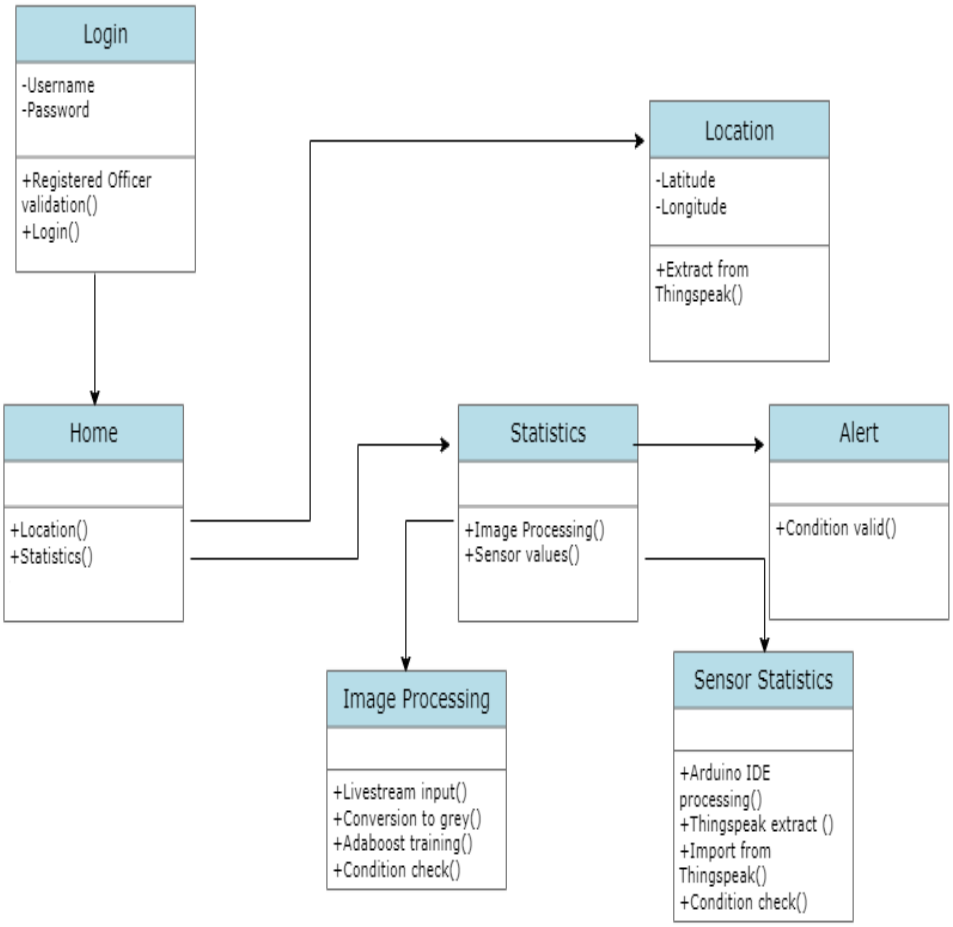


Figure 5.3: Usecase Diagram

The use case diagram depicted in Figure 5.3 shows the relation between the user and the system. The figure shows the relation between the Registered forest officer as actor and the relationship between the actor and app as use cases.This diagram is helpful in capturing the system's intended functionality and the roles of different actors involved.

5.3 Class Diagram

Figure 5.4: Class Diagram



The class diagram depicted in the Figure 5.4 shows the class diagram with the various pages which have various attributes and methods associated with the fire detection system.The diagram showcases the overall design of the system, highlighting the classes and their interactions. It helps in understanding the system's architecture, identifying key components, and facilitating communication among development teams.

Chapter 6

Implementation

The implementation phase brings the forest fire detection system to life, integrating hardware and software modules . Rigorous testing ensures reliability and accuracy. This report provides insights into the process, challenges, and outcomes of the implementation.

6.1 Tools Used for Designing Web Application Interface

6.1.1 Flutter

The web interface in the provided code was designed using the Flutter framework, which is an open-source UI toolkit developed by Google. Flutter allows for cross-platform development, enabling the creation of web, mobile, and desktop applications from a single codebase.To build the web interface, the Flutter framework leverages the Material Design system, which provides a set of visual guidelines and components for creating modern and visually appealing user interfaces. The Material Design system ensures consistency and usability across different devices and platforms.The software was used on the detailed analysis provided in publication by Hussain *et al.*[6].

6.1.2 Android Studio

Android Studio is the official Integrated Development Environment (IDE) for Android app development. It offers a rich set of tools and features, including a powerful code editor with intelligent code completion, a layout editor for visual interface design, and the Gradle build system for managing dependencies and packaging apps. With its emulator and device manager, developers can easily test their apps on virtual or physical devices. The IDE provides debugging and profiling tools, seamless version control integration, and support for libraries and APIs. It is highly extensible through plugins and offers a comprehensive development environment for building high-quality Android applications.The software was used on the detailed analysis provided in publication by J.DiMarzio*et al.*[7].

6.1.3 Visual Studio Code

Visual Studio Code is a lightweight and versatile source code editor developed by Microsoft, widely used for Android app development. With its intuitive interface, extensive plugin ecosystem, and features like syntax highlighting and debugging tools, it provides an efficient and customizable development environment. It supports Git integration, build systems, and task runners, making it a popular choice among developers.The software was used on the detailed analysis provided in publication by S.Aman*et al.*[8].

6.2 Tools Used for Designing Hardware Interface

6.2.1 Android IDE:

It is the official integrated development environment (IDE) for programming Arduino boards. Arduino IDE provides a user-friendly interface for writing, compiling, and uploading code to the ESP32 board. It supports various libraries and simplifies the process of hardware configuration.The software was used on the detailed analysis provided in Y.A.Badamasi[9].

6.2.2 ESP32 Board:

ESP32 is a widely used microcontroller board that provides Wi-Fi and Bluetooth connectivity. It is popular for IoT applications and can be programmed using Arduino IDE. The ESP32 board acts as the main hardware platform for connecting and interacting with sensors.The hardware was used on the detailed analysis provided in A.Maier*et al.*[10].

6.2.3 ThingSpeak:

Thingspeak is an IoT analytics platform and cloud service provided by MathWorks. It allows users to collect, store, analyze, and visualize data from IoT devices or sensors. With Thingspeak, you can create IoT applications and build dashboards to monitor and interact with your data. It supports various protocols for data communication, including HTTP and MQTT.he software was used on the detailed analysis provided in publication by Dr.SRajaprakash *et al.*[11].

6.3 Tools Used for Image Processsing

6.3.1 Python:

Python is a popular high-level programming language known for its simplicity and readability.Python is widely used in the field of computer vision due to its rich ecosystem of libraries and frameworks specifically designed for image processing and computer vision tasks. The most prominent library in this domain is OpenCV (Open Source Computer Vision Library), which provides a comprehensive set of tools and functions for image and video analysis, object detection and tracking, image enhancement, and more. Python's simplicity and readability make it easy to work with complex computer vision algorithms and techniques, allowing developers to quickly prototype and implement various vision-based applications.

Chapter 7

Results and Discussion

The results obtained and objectives accomplished throughout the development of the project is described here.All the basic requirements required for a warning and detection of forest fire are met.The objectives discussed at the beginning of the project are met with maximum efficiency.

7.1 Results and Discussion

The project aimed in developing a warning system for forest fire which was intended to deliver the following objectives.

- To detect Forest Fire in Livestream feeded by an IP Camera.
- To detect Forest Fire using IOT via sensors
- To create an app to feed and analyze the Data provided from Livestream and IOT which can be used by the authorities to warn about forest fire without full-time surveillance
- Device an alarm buzzer to prompt the authorities at right time by the signals send by the app

7.2 Screenshots

The implemented forest fire detection system consists of several key components and functionalities. Figure 7.1 showcases the login page, accessible only to registered forest officers with unique username and login ID. Figure 7.2 presents the home page, offering services like location tracking, log file access, and statistics. The location page, depicted in Figure 7.3, displays the latitude and longitude coordinates of the hardware sensor modules, obtained through the integrated ThingSpeak service. Figure 7.4 represents the statistics page, where feeds are only visible after pressing the start detection button.

Figures 7.5 and 7.6 exhibit the setup of IP livestream and IP camera on a mobile device. In Figure 7.7, the IP stream captures the occurrence of a fire, triggering the detection system. Figure 7.8 provides an overview of the hardware components utilized, while Figure 7.9 showcases the initiation of smoke detection. When smoke is detected, Figure 7.10 displays a message indicating smoke detection, along with the corresponding smoke and temperature values, and an image confirming fire detection. At this point, an alarm sound is activated, meeting the required conditions for alarming. Lastly, Figure 7.11 presents the log page, featuring statistical graphs of temperature and smoke levels plotted against dates, as well as an image depicting the location. Together, these visual representations demonstrate the comprehensive functionality and effectiveness of the forest fire detection system.

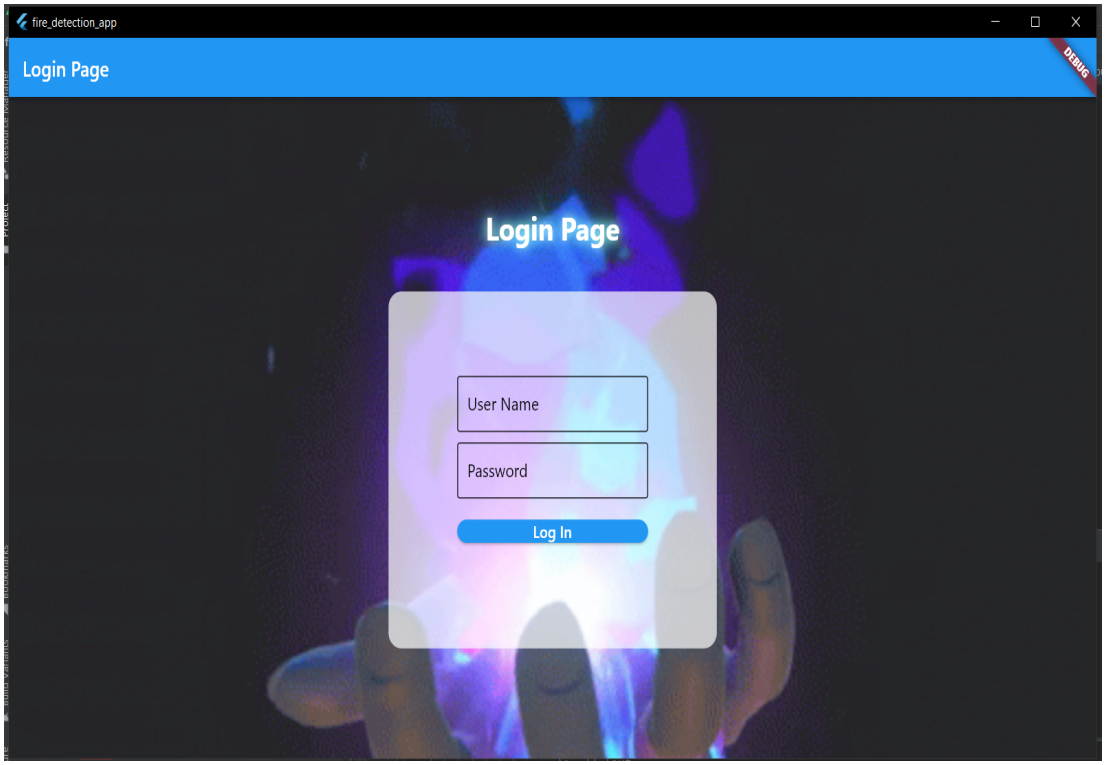


Figure 7.1: login page

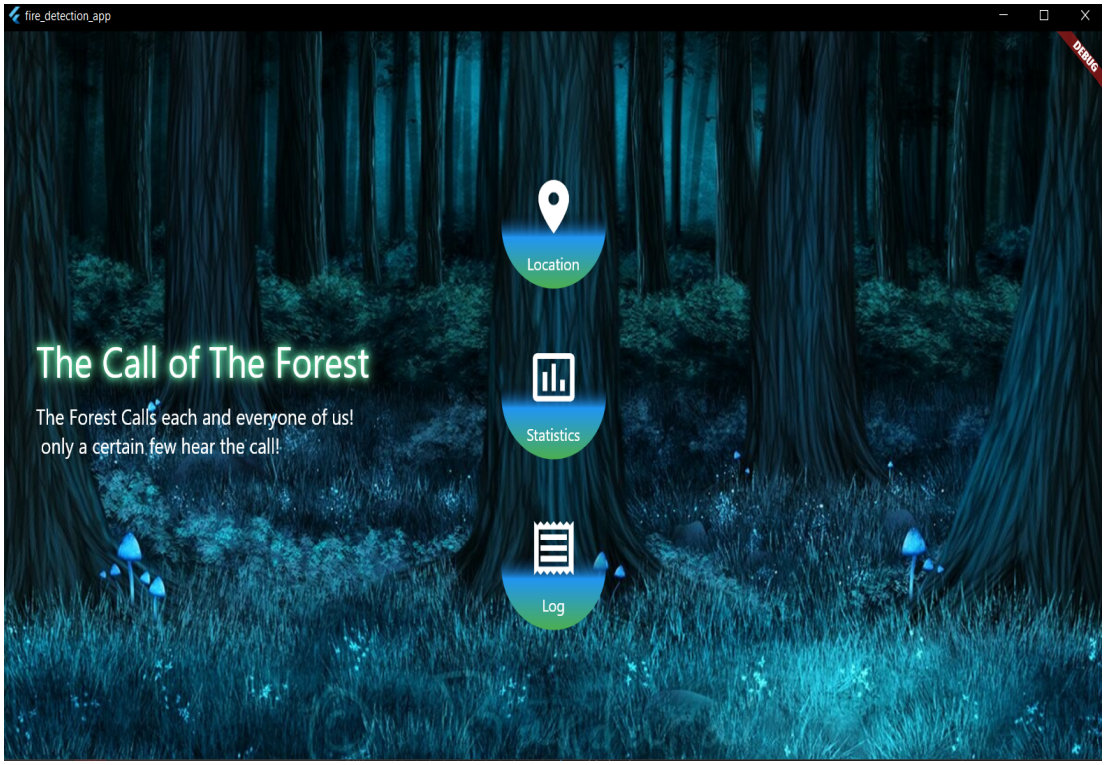


Figure 7.2: Home Page



Figure 7.3: Location Page

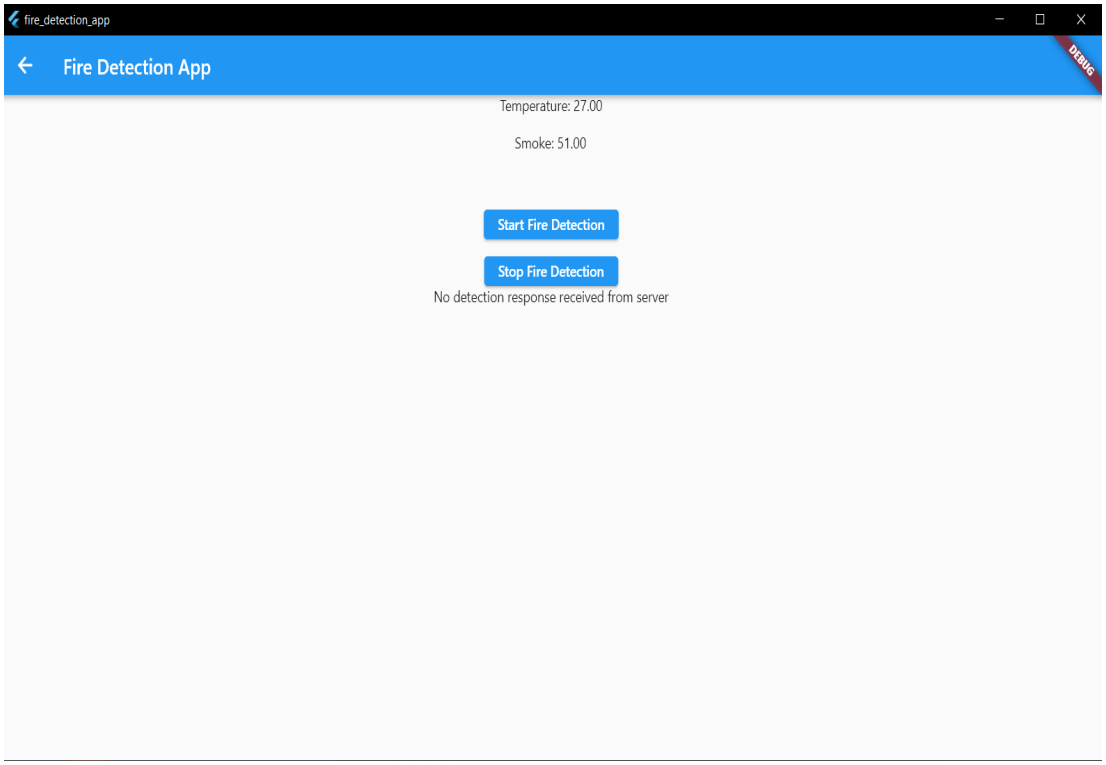


Figure 7.4: Statistics Page

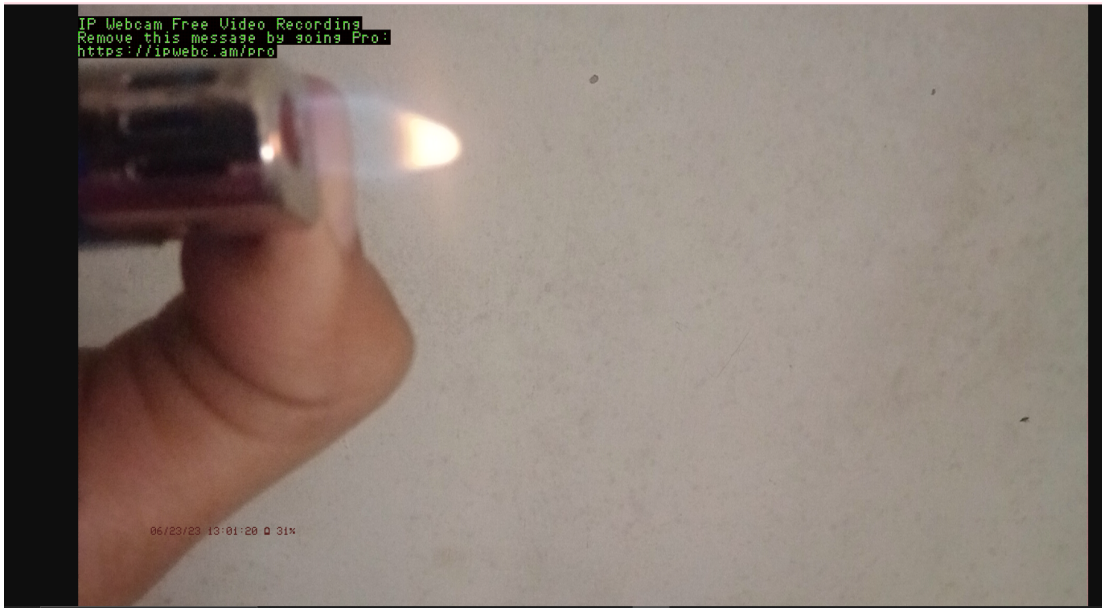


Figure 7.5: Livefeed from IP Camera



Figure 7.6: Mobile IP Camera

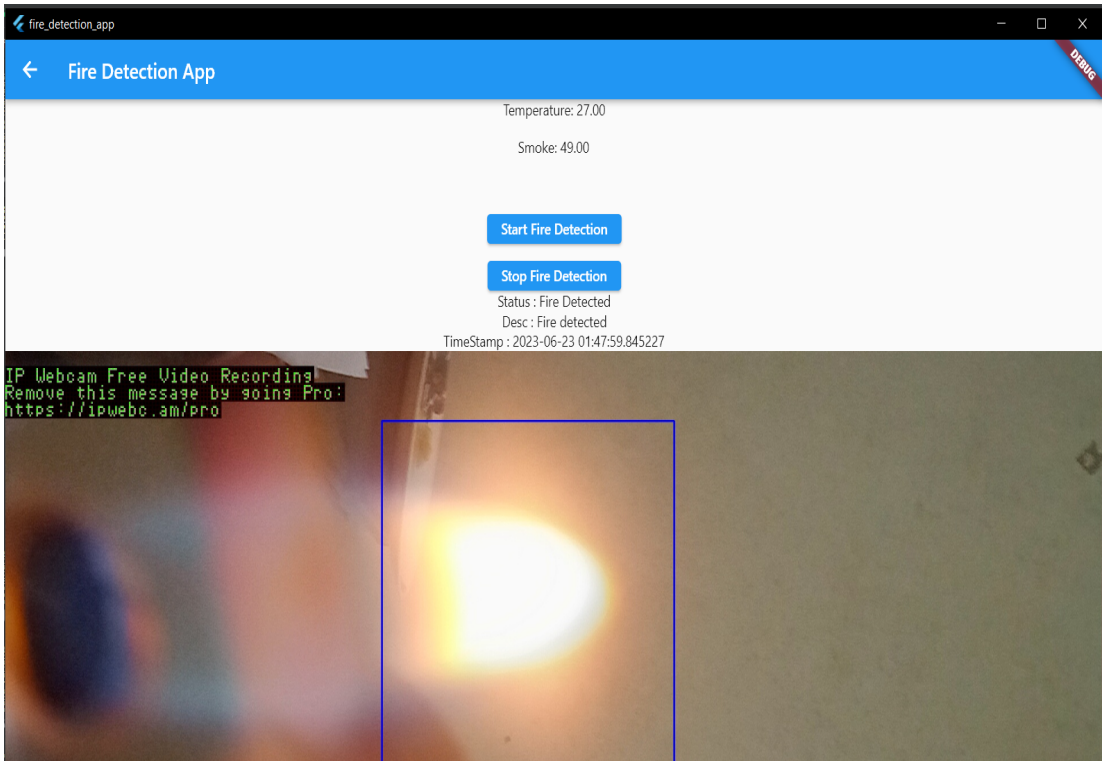


Figure 7.7: Fire Detected From IP Stream

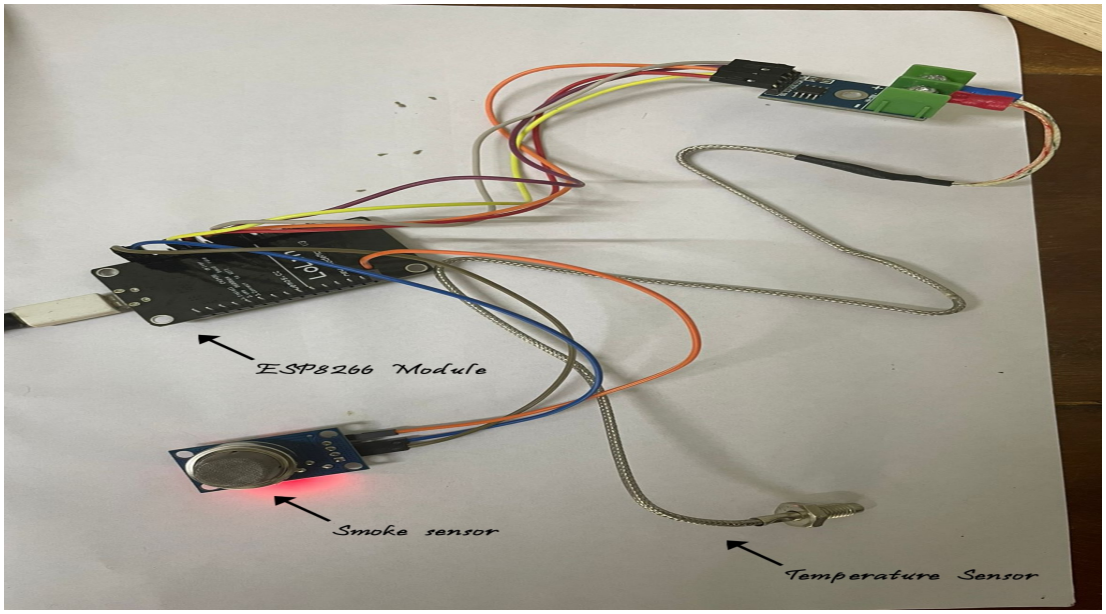


Figure 7.8: Hardware Components

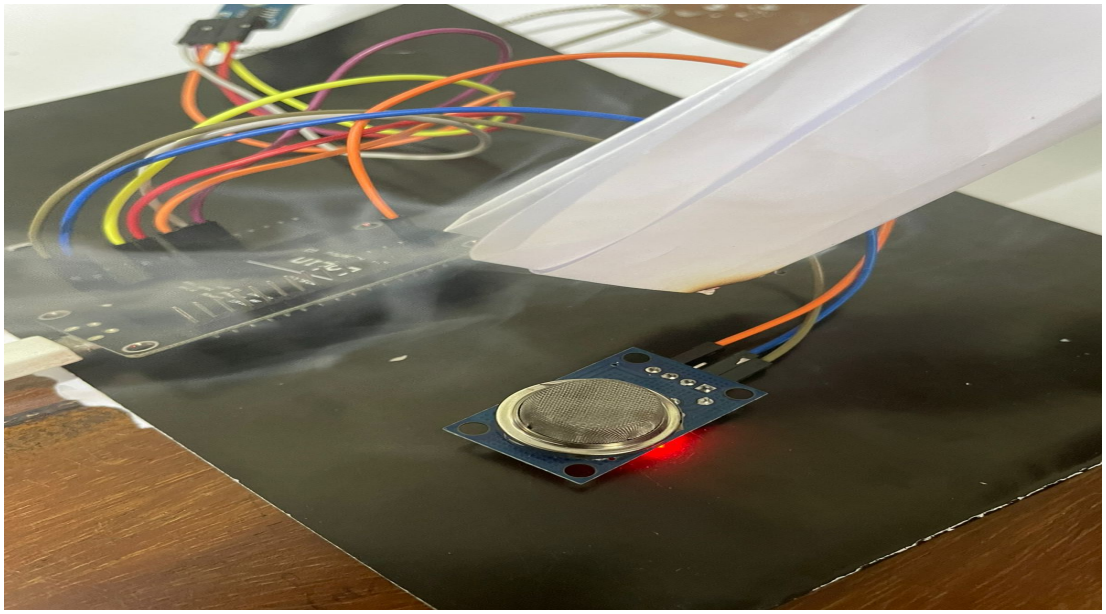


Figure 7.9: Smoke initiated

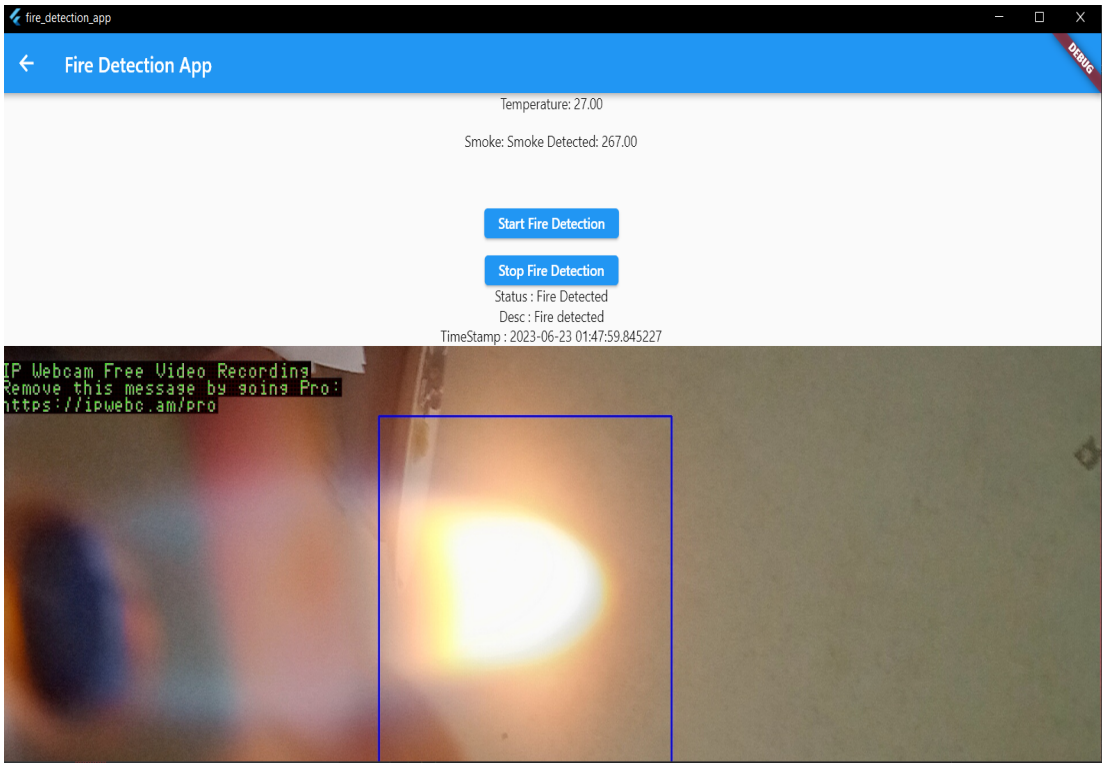


Figure 7.10: Fire Detected From IP Stream and Smoke Sensor

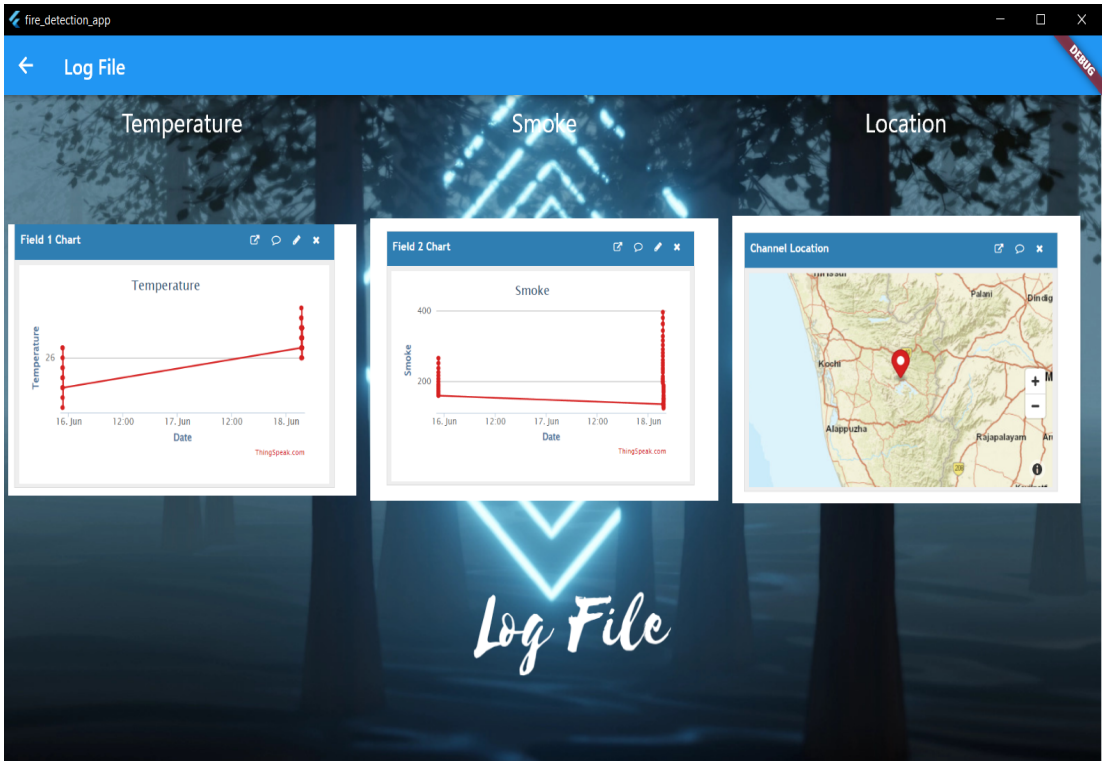


Figure 7.11: Log Page

Chapter 8

Testing

The chapter different testing's are done inorder to ensure the proper functioning of the application. It is the practice of making objective judgments regarding the extent to which the. system meets, exceeds or fails to meet stated objectives.

8.1 Functionality Testing

For each module discussed in the functional description of Implementation of Image Processing and IoT Implementation the following are verified.

- verified Sensor detection module
- verified Fire detection module
- Verified location module
- Verified Log system
- verified alerting

8.2 UI Testing

The forest fire detection application underwent thorough UI testing to ensure its functionality and user experience.

- Comprehensive UI testing was conducted on the forest fire detection application.All UI components, including buttons, text fields, and images, functioned correctly.
- Visual testing ensured that background images, container transparency, and text styles were accurately displayed.
- Simulated user interactions were performed to evaluate the app's responsiveness.

8.3 Integration Testing

Integration testing for the Fire Detection Screen verifies the collaboration and proper functioning of different components and modules within the application to ensure they work together smoothly and produce the desired results.

- Fetch Data from API : The data is fetched and displayed correctly, including temperature and smoke values.
- Alert Message : When the smoke level and exceeds the threshold and fire frames are detected , the alert message is shown.
- Start Fire Detection : The test validates the ability to start fire detection by making a POST request to the server.Fire detection should start successfully. Fire detection starts as expected, and the UI updates to show that the detection is active.
- Real-time Data and Image Display: The test verifies the real-time data and image display functionality using a WebSocket connection. The screen shows the real-time detection status, description, timestamp.

8.4 Network Testing

The network testing for the Fire Detection Screen validates the functionality and reliability of networking components, including API communication, WebSocket connections, and error handling.

- API Data Retrieval and API Response Handling : The HTTP GET request successfully fetch data from the API.The application handles success, error, and empty responses correctly, updating the UI accordingly.
- WebSocket Connection :The WebSocket connection is established successfully, and data is received from the server.
- Error Handling :The application handles network errors properly, displaying error messages and handling exceptions.

Chapter 9

Conclusion

The current state of forest fire detection technologies suffers from limitations such as false detection, delayed warnings, and unresolved issues, often relying on large datasets or unreliable hardware. These shortcomings can lead to devastating consequences, including extensive forest destruction and inefficient resource management.

In response, our proposed solution tackles false detection while providing precise, location-specific, and timely results. By integrating live stream image processing and sensor data through a web app, we can accurately detect forest fires and generate alarms promptly.

Leveraging machine learning algorithms and sensor responses, our approach minimizes false detection occurrences, enhancing the efficiency and reliability of forest fire detection systems. This advancement empowers forest officers to respond swiftly and effectively, safeguarding natural resources and mitigating catastrophic events.

References

[1] J. Marin K and A. S Kannan, ”Review on Fire Detection Techniques” *Electronic Journal*,pp.2, 2020, doi: https://doi.org/10.2139/ssrn.3794284

[2] X. Wu, X. Lu, and H. Leung , A Video Based Fire Smoke Detection Using Robust AdaBoost,*Sensors*,pp.2 , Nov. 2018, doi: 10.3390/s18113780.

[3] an, Xian-Feng Jin, Jesse Wang, Ming-Jie Jiang, Wei Gao, Lei Xiao, Li-Ping, “Video fire detection based on Gaussian Mixture Model and multi-color features” *Signal Image Video Process*,pp.4, 2017. doi:10.1007/s11760-017-1102-y.

[4] F. Khan , Xu Z, Sun J, Khan FM, Ahmed A, Zhao Y.“Recent Advances in Sensors for Fire Detection” *Sensors (Basel)*,pp.4 , 2022. doi:10.3390/s22093310.

[5] S. Mehtab and J. Sen, *Object Detection and Tracking Using OpenCV in Python,*,pp.6, 2020. doi:10.13140/RG.2.2.34682.93128.

[6] Hussain, Hina khan, Kamran Farooqui, Faiza Faiza Ali, Qasim Siddiqui, Isma. ”Comparative Study of Android Native and Flutter App Development”,pp.19. 2021,

[7] J. DiMarzio, ”Beginning Android Programming with Android Studio,” John Wiley Sons, pp.19,2016.

[8] S. Amann, S. Proksch, S. Nadi and M. Mezini, ”A Study of Visual Studio Usage in Practice,” 2016 *IEEE 23rd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*, Osaka, Japan, 2016, pp.20, doi: 10.1109/SANER.2016.39.

[9] Y. A. Badamasi, ”The working principle of an Arduino,” *2014 11th International Conference on Electronics, Computer and Computation (ICECCO)*, Abuja, Nigeria, 2014, pp.20, doi: 10.1109/ICECCO.2014.6997578.

[10] A. Maier, A. J. Sharp, and Y. Vagapov, *Comparative analysis and practical implementation of the ESP32 microcontroller module for the internet of things*. 2017,pp.20, doi: 10.1109/itecha.2017.8101926.

[11] Dr.SRajaprakash ,B. Kasthuriraja, “Weather Analysis Using Thingspeak,” *ijirt.org*, Sep. 02, 2020,pp.21 https://ijirt.org/Article?manuscript=150203